

Module3_Lab3

September 19, 2026

Firstly, I'm running the four commands for this new bridges.csv dataset

```
import pandas as pd
df = pd.read_csv("bridges.csv")
df.head()
```

This gives me a preview of the first five rows of the dataset.

```
[2]: df.shape()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[2], line 1
----> 1 df.shape()

TypeError: 'tuple' object is not callable
```

```
[3]: df.shape
```

```
[3]: (108, 13)
```

This tells me that there are 108 rows and 13 columns.

```
[4]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 108 entries, 0 to 107
Data columns (total 13 columns):
 #   Column      Non-Null Count  Dtype  
---  -
 0   Id          108 non-null   str     
 1   river       108 non-null   str     
 2   location    107 non-null   float64 
 3   erected     108 non-null   int64   
 4   purpose     108 non-null   str     
 5   length      81 non-null    float64 
 6   lanes       92 non-null    float64 
 7   clear-g     106 non-null   str     
 8   t-or-d      102 non-null   str     
 9   material    106 non-null   str     
10  span        92 non-null    str
```

```
11 rel-l      103 non-null    str
12 type       106 non-null    str
dtypes: float64(3), int64(1), str(9)
memory usage: 14.7 KB
```

```
[5]: df.describe()
```

```
[5]:
```

	location	erected	length	lanes
count	107.000000	108.000000	81.000000	92.000000
mean	25.978505	1905.314815	1567.469136	2.630435
std	13.665871	37.173664	747.491523	1.164657
min	1.000000	1818.000000	804.000000	1.000000
25%	15.500000	1884.000000	1000.000000	2.000000
50%	27.000000	1903.000000	1300.000000	2.000000
75%	37.500000	1928.000000	2000.000000	4.000000
max	52.000000	1986.000000	4558.000000	6.000000

```
[6]: df.isnull().sum()
```

```
[6]: Id          0
river          0
location       1
erected        0
purpose        0
length        27
lanes         16
clear-g        2
t-or-d         6
material        2
span          16
rel-l          5
type           2
dtype: int64
```

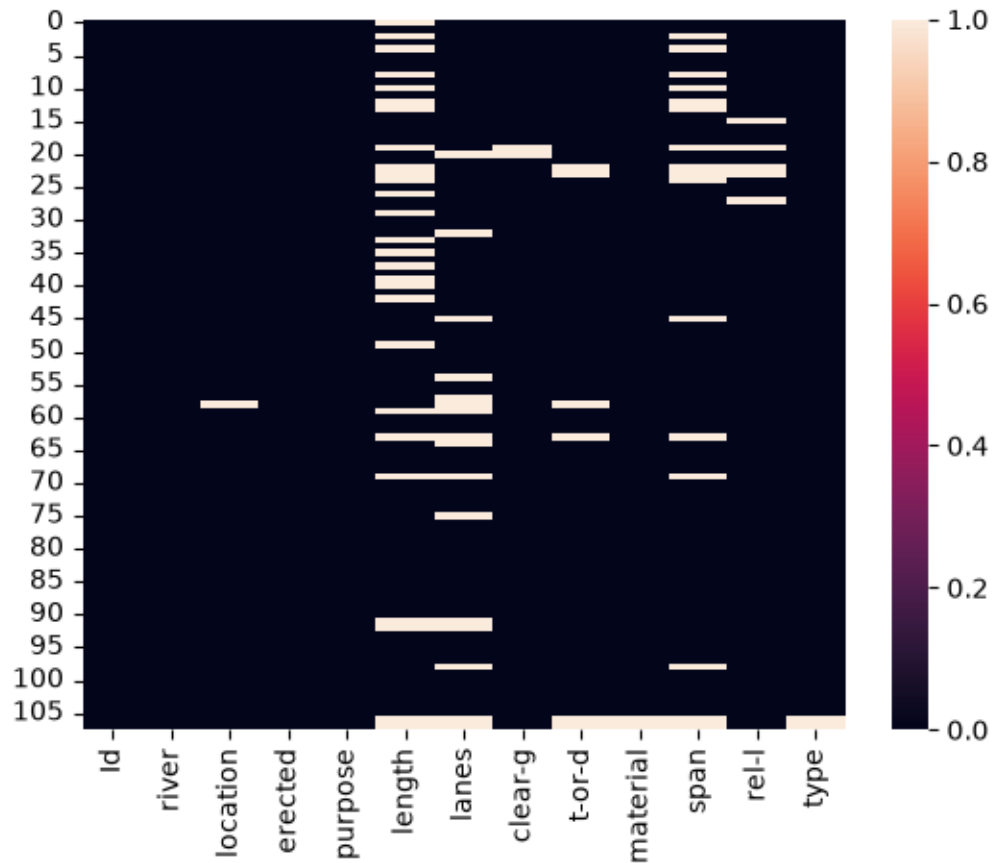
Using the `isnull().sum()`, we see that there are a few missing values (1 location, 27 length, 16 lanes, 2 clear-g, 6 t or d, 2 material, 16 span, 5 rel-l, and 2 type).

```
[7]: import pandas as pd
import seaborn as sns

df = pd.read_csv("bridges.csv")

sns.heatmap(df.isnull())
```

```
[7]: <Axes: >
```



On this seaborn heatmap, it appears that for row 105, there are a number of horizontal bands that tell us that there are missing data across 5 of the columns (length, lanes, t-or-d, material, span and type). All other rows have only at most 2-3 missing values in the columns. Therefore this tells us that it's an extreme outlier in a sense due to a number of records that is incomplete.

Also another one is between row 55-60, on the location column, it is the only missing value versus all others.

[]: